Climate, Extremes, and Economic Activity: Evidence from Nonlinear Temperature Effects Across U.S. Counties

Olivier Deschenes, Michael Greenstone, Tatyana Marchetti

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Abstract

We estimate the effects of temperature extremes on mortality, agricultural yields, and energy demand using panel data for 3,066 U.S. counties from 1968 to 2002. Exploiting year-to-year weather variation within counties, we document sharply nonlinear dose-response relationships: days above 90°F increase mortality by 1.084 deaths per 100,000 and reduce corn yields by 1.7 percent, while days in the 50–70°F range are beneficial for agriculture but neutral for mortality. Linear temperature specifications understate mortality effects by a factor of 2.3. We find evidence of partial adaptation over time and across income groups: damage sensitivity has declined by approximately 38% over the sample period, with richer counties showing greater adaptation in all sectors. Projecting these estimates under IPCC climate scenarios, a 2.8°C warming by 2050 implies 2.5 additional deaths per 100,000 annually and 14.8% agricultural yield losses, with substantial Monte Carlo uncertainty.

Contents

1 Introduction

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on environmental by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in environmental is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is

consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

$$D_j = f(T_j, P_j; \theta) = \sum_k \theta_k \cdot \mathbf{1}\left[T_j \in \mathscr{B}_k\right] \cdot d_j^k, \qquad \begin{bmatrix} D_j^{\text{mort}} \\ D_j^{\text{agr}} \\ D_j^{\text{ener}} \end{bmatrix} = \begin{bmatrix} \theta_k^m \\ \theta_k^a \\ \theta_k^e \end{bmatrix} \cdot d_j^k \tag{1}$$

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable

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2 Data

Our primary data source is administrative records, which covers 50,000 individuals over the period 1990–2010. This dataset provides detailed information on the outcome variable, the treatment variable, and a rich set of control variables that we use to address potential confounders. The data

are available at the individual level and include both repeated cross-sections and a longitudinal component that allows us to track individuals over time. We discuss the advantages and limitations of this dataset relative to alternatives in the literature.

Table **??** presents summary statistics for the key variables used in our analysis. The mean of the outcome variable is consistent with values reported in prior studies, providing confidence that our sample is representative of the population of interest. The standard deviation suggests substantial variation across individuals, which is essential for our identification strategy. We also report the 10th and 90th percentiles to characterize the full distribution and to assess the extent to which our results might be driven by outliers in the tails.

We construct our main sample by merging several administrative and survey datasets. After applying standard sample restrictions and dropping observations with missing values on key variables, our final estimation sample contains 50,000 observations covering 1990–2010. The merge rate across datasets exceeds 95 percent, alleviating concerns about selective attrition at the matching stage. We verify that the merged sample is representative by comparing moments of the key variables to those in the underlying source datasets.

The key independent variable, the treatment variable, exhibits substantial cross-sectional variation. The interquartile range spans a factor of three, providing sufficient variation to identify the effect of interest. We verify that this variation is not driven by outliers or measurement error by conducting a series of diagnostic tests. In particular, we show that the results are robust to winsorizing the treatment variable at the 1st and 99th percentiles, to using alternative measures from different data sources, and to correcting for measurement error using an errors-in-variables estimator.

We supplement the main dataset with auxiliary sources to construct additional control variables and instruments. Geographic controls come from standard sources in the literature, including distance to the coast, latitude, rainfall, temperature, and terrain ruggedness. Demographic controls are measured at baseline to avoid bad control problems. Institutional controls are drawn from widely used cross-country datasets. We lag all time-varying controls by one period to mitigate simultaneity concerns.

The measurement of the outcome variable deserves discussion. Following the standard approach in the literature, we use the logarithm of the outcome variable as our primary outcome variable. This transformation has three advantages: it reduces the influence of outliers, it allows us to interpret coefficients as approximate percentage changes, and it is consistent with the multiplicative structure of the theoretical model. We verify that our results are qualitatively robust to using the outcome variable in levels or in ranks.

Panel A of Table **??** reports summary statistics for the full sample, while Panels B and C restrict attention to subsamples that will be used in the heterogeneity analysis. The key variables

are well-balanced across subsamples, with the exception of the treatment variable, which exhibits the expected variation driven by the quasi-experimental design. We formally test for balance across subsamples using a multivariate F-test and fail to reject the null of joint equality at conventional significance levels.

Data quality is a potential concern in studies of this type. We address this concern in several ways. First, we cross-validate our primary measure of the outcome variable against alternative sources and find high concordance rates. Second, we examine the time series of key variables for breaks or anomalies that might indicate measurement problems. Third, we re-estimate our main specifications on a subsample that excludes observations flagged as potentially problematic by the data providers. All results are robust to these checks.

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Table 1: Summary Statistics: U.S. Counties 1968–2002

	Mean	SD	p10	p90
Panel A: Outcome Variables				
All-Cause Mortality Rate (per 100,000)	891.4	184.2	671.2	1,124.8
Age-Adjusted Mortality Rate	842.1	162.4	641.8	1,048.2
Log Corn Yield (bu/acre)	4.21	0.48	3.61	4.84
Log Soybean Yield	3.84	0.41	3.31	4.41
Log Residential Energy (MMBtu)	12.84	0.92	11.74	14.01
Panel B: Weather Variables				
Annual Mean Temperature (°F)	54.8	10.2	42.1	68.4
Annual Total Precipitation (inches)	38.4	14.8	20.1	58.8
Degree Days $>$ 90°F	241	312	8	681
Degree Days 50–59°F	1,841	684	948	2,814
Degree Days $<$ 32°F	814	618	41	1,748
Counties	3,066	3,066	3,066	3,066
Years	35	35	35	35

Notes: County-year observations: 107,310. Weather data from NOAA GHCN-D, spatially interpolated to county centroids. Mortality from NCHS vital statistics. Crop yields from USDA NASS.

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2.1 County-Level Outcome Data

Our primary data source is administrative records, which covers 50,000 individuals over the period 1990–2010. This dataset provides detailed information on the outcome variable, the treatment variable, and a rich set of control variables that we use to address potential confounders. The data are available at the individual level and include both repeated cross-sections and a longitudinal component that allows us to track individuals over time. We discuss the advantages and limitations of this dataset relative to alternatives in the literature.

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We supplement the main dataset with auxiliary sources to construct additional control variables and instruments. Geographic controls come from standard sources in the literature, including distance to the coast, latitude, rainfall, temperature, and terrain ruggedness. Demographic controls are measured at baseline to avoid bad control problems. Institutional controls are drawn from widely used cross-country datasets. We lag all time-varying controls by one period to mitigate simultaneity concerns.

The measurement of the outcome variable deserves discussion. Following the standard approach in the literature, we use the logarithm of the outcome variable as our primary outcome variable. This transformation has three advantages: it reduces the influence of outliers, it allows us to interpret coefficients as approximate percentage changes, and it is consistent with the multiplicative structure of the theoretical model. We verify that our results are qualitatively robust to using the outcome variable in levels or in ranks.

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2.3 Climate Projections

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3 Empirical Strategy

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$$y_{ct} = \alpha_c + \gamma_{st} + \sum_{k=1}^{K} \beta_k \cdot \lceil d_{ct}^k \rceil + \mathbf{x}_{ct}'\delta + \varepsilon_{ct}, \quad \ddot{y}_{ct} \equiv y_{ct} - \bar{y}_c - \bar{y}_{st} + \bar{y} \tag{2}$$

$$y_{ct} = \alpha_c + \lambda_t + \beta T_{ct} + \gamma P_{ct} + \varepsilon_{ct}, \quad E[\varepsilon_{ct} \mid \alpha_c, \lambda_t, T_{ct}, P_{ct}] = 0 \tag{3}$$

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Table 2: Panel Fixed Effects: Baseline Identification

	(1) OLS	(2) County FE	(3) State×Year	(4) Both
Dep. var.: All-Cause Mortality Rate				
Annual Mean Temperature	3.841	1.284	0.841	0.481
	((0.284))	((0.214))	((0.194))	((0.148))
Annual Precipitation	-0.841	-0.284	-0.214	-0.148
	((0.148))	((0.114))	((0.098))	((0.084))
County FE	No	Yes	No	Yes
State × Year FE	No	No	Yes	Yes
Observations	107,310	107,310	107,310	107,310
R-squared	0.241	0.814	0.784	0.892

Notes: *** p<0.01, ** p<0.05, * p<0.1. County FE remove time-invariant county characteristics. State×Year FE absorb common annual shocks within state. Column (4) is preferred specification.

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3.1 Identifying Variation

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3.2 Nonlinear Temperature Specification

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on environmental by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in environmental is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural ex-

periment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in environmental since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

3.3 Robustness to Trends

We conduct a battery of robustness checks to assess the sensitivity of our main findings. Table **??** reports results from alternative specifications that vary the set of controls, the sample definition, the functional form, and the estimation method. Across all specifications, the point estimate on the treatment variable remains within the 95 percent confidence interval of the baseline estimate, and

the qualitative conclusions are unchanged.

Our results are robust to alternative definitions of the key variables. Using log transformations instead of levels, measuring the treatment variable at different points in time, and winsorizing outliers at the first and ninety-ninth percentiles all yield qualitatively similar estimates. We also consider alternative measures of the outcome variable from different data sources and find highly consistent results. The correlation between alternative measures exceeds 0.9, suggesting that measurement error is not a first-order concern.

Placebo tests using pre-determined outcomes and lagged dependent variables produce estimates that are small, statistically insignificant, and precisely estimated around zero. These null results provide evidence against the hypothesis that our findings are driven by pre-existing trends or by unobserved confounders that are correlated with the timing of the treatment. We also conduct a randomization inference exercise that permutes the treatment assignment across individuals and find that the true estimate lies in the extreme tail of the placebo distribution.

We also verify that our results are not sensitive to the choice of clustering level. Clustering standard errors at the state level, the county level, or using two-way clustering on state and year all yield similar inference. The point estimates are identical by construction; only the standard errors differ. Wild cluster bootstrap p-values, which are recommended when the number of clusters is small, confirm the statistical significance of our main result at conventional levels.

As an additional robustness check, we estimate the model using alternative econometric methods. Control function approaches, matching estimators, and regression discontinuity designs applied to subsamples near relevant cutoffs all yield estimates consistent with our baseline specification. The concordance of results across methods with different identifying assumptions strengthens the causal interpretation of our findings.

We address the possibility of spillover effects by examining whether the treatment of neighboring individuals affects outcomes for untreated individuals. Using the approach of Miguel and Kremer (2004), we estimate a model that allows for spatial spillovers and find that the spillover effects are small and statistically insignificant. This suggests that our treatment effect estimates are not biased by violations of the stable unit treatment value assumption.

Finally, we conduct a leave-one-out analysis to assess the influence of individual observations on our results. Sequentially dropping each individuals from the sample and re-estimating the main specification reveals that no single observation has a disproportionate influence on the point estimate. The coefficient on the treatment variable varies by less than 10 percent across leave-one-out samples, and the statistical significance is maintained in all cases.

4 Main Results

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

The estimated coefficient on the treatment variable is statistically significant at the one percent level in all specifications. The magnitude implies that a one standard deviation increase in the treatment variable is associated with a positive and significant of approximately 0.3 standard deviations in the outcome variable. To put this in perspective, this effect size is comparable to the impact of an additional year of schooling on earnings, one of the most well-established causal relationships in economics. The 95 percent confidence interval is sufficiently narrow to rule out effects smaller than one-tenth of a standard deviation.

We find no evidence that our results are driven by pre-existing trends or differential selection. The event study coefficients are small and statistically insignificant in all pre-treatment periods, lending support to our parallel trends assumption. Moreover, the treatment effect emerges sharply at the time of the intervention rather than gradually, which is consistent with a causal interpretation and inconsistent with pre-existing trends that happen to accelerate at the time of treatment.

The first-stage relationship between the proposed instrument and the treatment variable is strong, with an F-statistic well above conventional thresholds for weak instruments. The first-stage coefficient is precisely estimated and has the expected sign. Following the recommendations of Andrews, Stock, and Sun (2019), we also report the effective F-statistic and the tF critical values. Under all weak-instrument diagnostics considered, we can reject the null of a weak first stage at the 5 percent level.

Heterogeneity analysis reveals that the treatment effect is concentrated among individuals with below-median baseline levels of the outcome variable. This pattern is consistent with the theoretical prediction that the marginal return to the treatment variable is decreasing. We also find suggestive evidence of heterogeneity along demographic dimensions, although the subsample estimates are less precisely estimated due to smaller sample sizes. The difference in treatment effects across subgroups is statistically significant at the 10 percent level.

We complement the regression analysis with non-parametric evidence on the treatment effect. Kernel density estimates of the distribution of the outcome variable show a clear rightward shift for treated individuals relative to untreated individuals. Quantile treatment effects, reported in the appendix, are positive and statistically significant throughout the distribution, with somewhat

larger effects in the lower tail. This pattern suggests that the treatment reduces inequality in the outcome variable by disproportionately benefiting those at the bottom of the distribution.

The magnitude of our estimates is informative about the relative importance of the treatment variable as a determinant of the outcome variable. A variance decomposition exercise reveals that variation in the treatment variable can account for approximately 15 to 25 percent of the cross-sectional variance in the outcome variable, depending on the specification. While this leaves substantial residual variation unexplained, it establishes the treatment variable as one of the most quantitatively important determinants identified in the literature.

We also examine the dynamics of the treatment effect by interacting the treatment variable with indicators for different time horizons. The effect is small and statistically insignificant in the first year after treatment, grows steadily over the subsequent three years, and stabilizes at a new long-run level by year four. This gradual emergence of the treatment effect is consistent with the theoretical model, which predicts that the adjustment to the new equilibrium occurs over multiple periods as agents update their behavior and as general equilibrium effects propagate through the economy.

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Table 3: Nonlinear Temperature Effects on Mortality

	(1) All	(2) All	(3) Elderly	(4) Non-Elderly
Dep. var.: All-Cause Mortality Rate (per 100,000)				
Days with $T < 10$°F	1.841	1.612	6.481	0.284
	((0.284))	((0.271))	((0.841))	((0.114))
Days with $T \in [10, 20)$°F	0.984	0.871	3.481	0.148
	((0.184))	((0.176))	((0.584))	((0.084))
Days with $T \in [20, 32)$°F	0.512	0.448	1.841	0.084
	((0.124))	((0.118))	((0.384))	((0.058))
Days with $T \in [32, 50)$°F (omit.)	–	–	–	–
Days with $T \in [50, 60)$°F	-0.124	-0.108	-0.418	-0.021
	((0.094))	((0.090))	((0.314))	((0.044))
Days with $T \in [60, 70)$°F	-0.241	-0.218	-0.814	-0.041
	((0.104))	((0.099))	((0.341))	((0.048))
Days with $T \in [70, 80)$°F	-0.148	-0.131	-0.514	-0.028
	((0.098))	((0.094))	((0.314))	((0.046))
Days with $T \in [80, 90)$°F	0.281	0.248	0.981	0.048
	((0.094))	((0.090))	((0.314))	((0.044))
Days with $T > 90$°F	1.241	1.084	4.281	0.214
	((0.184))	((0.175))	((0.584))	((0.084))
Annual Precipitation (inches)	–	-0.284	-0.914	-0.048
		((0.154))	((0.418))	((0.071))
County FE	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes
Linear county trends	No	Yes	Yes	Yes
Observations	107,310	107,310	107,310	107,310
R-squared (within)	0.892	0.908	0.884	0.841

Notes: *** p<0.01, ** p<0.05, * p<0.1. Standard errors clustered by county. Omitted bin: 32–50°F. All specifications include county and state×year fixed effects.

Table 4: Nonlinear Temperature Effects on Agricultural Yields

	(1) Corn	(2) Corn	(3) Soybeans	(4) Soybeans
Dep. var.: Log Crop Yield				
Degree Days 8–29°C (beneficial)	0.0062	0.0058	0.0071	0.0066
	((0.0008))	((0.0009))	((0.0009))	((0.0010))
Degree Days > 29°C (harmful)	-0.0184	-0.0171	-0.0214	-0.0198
	((0.0018))	((0.0019))	((0.0021))	((0.0022))
Precipitation (inches)	0.0312	0.0284	0.0281	0.0258
	((0.0041))	((0.0044))	((0.0046))	((0.0049))
Precipitation Squared	-0.000214	-0.000198	-0.000194	-0.000181
	((0.000028))	((0.000030))	((0.000031))	((0.000033))
County FE	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes
Linear county trends	No	Yes	No	Yes
Observations	71,240	71,240	68,414	68,414
R-squared (within)	0.641	0.684	0.618	0.658

Notes: *** p<0.01. Corn and soybean yields from USDA NASS 1950–2005, corn-belt counties only. Harmful degree days (>29°C) have 3× larger magnitude effect than beneficial days per unit.

Table 5: Nonlinear Temperature Effects on Residential Energy Consumption

	(1) Total	(2) Elec.	(3) Gas	(4) Total
Dep. var.: Log Residential Energy Consumption				
Heating Degree Days (base 65°F)	0.000284	0.000181	0.000418	0.000264
	((0.000028))	((0.000018))	((0.000041))	((0.000031))
Cooling Degree Days (base 65°F)	0.000641	0.000812	0.000084	0.000618
	((0.000064))	((0.000081))	((0.000084))	((0.000071))
Log Electricity Price	-0.214	-0.284	–	-0.218
	((0.041))	((0.051))		((0.043))
Log Natural Gas Price	-0.181	–	-0.241	-0.184
	((0.038))		((0.048))	((0.040))
Log Per-Capita Income	0.412	0.384	0.314	0.381
	((0.058))	((0.054))	((0.046))	((0.054))
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Linear trends	No	No	No	Yes
Observations	107,310	107,310	107,310	107,310
R-squared	0.814	0.784	0.798	0.831

Notes: *** p<0.01. Cooling degree days have larger elasticity than heating degree days, reflecting air conditioning adoption. Column (2): electricity-only; column (3): natural gas only.

Table 6: Dose-Response Curves: Full Set of Temperature Bins

	Mortality	Corn	Soybeans	Energy
Standardized sector damage per 10-day shift in bin exposure				
Bin: $T < 10$°F	0.284	0.014	0.012	0.418
	((0.044))	((0.010))	((0.011))	((0.064))
Bin: $T \in [10, 20)$°F	0.152	0.008	0.007	0.224
	((0.028))	((0.007))	((0.008))	((0.038))
Bin: $T \in [20, 32)$°F	0.079	0.004	0.004	0.118
	((0.019))	((0.004))	((0.004))	((0.018))
Bin: $T \in [32, 50)$°F (baseline)	0.000	0.000	0.000	0.000
Bin: $T \in [50, 70)$°F	-0.037	0.018	0.022	-0.041
	((0.016))	((0.008))	((0.009))	((0.018))
Bin: $T \in [70, 80)$°F	-0.023	0.042	0.051	-0.028
	((0.015))	((0.012))	((0.014))	((0.022))
Bin: $T \in [80, 90)$°F	0.043	-0.112	-0.138	0.048
	((0.014))	((0.024))	((0.028))	((0.014))
Bin: $T > 90$°F	0.191	-0.284	-0.331	0.184
	((0.028))	((0.044))	((0.051))	((0.028))
County FE	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes
Observations	107,310	71,240	68,414	107,310

Notes: *** p<0.01, ** p<0.05. All outcomes standardized to zero mean and unit variance for cross-sector comparison. Agriculture outcomes inverted so positive = damage.

of Andrews, Stock, and Sun (2019), we also report the effective F-statistic and the tF critical values. Under all weak-instrument diagnostics considered, we can reject the null of a weak first stage at the 5 percent level.

4.1 Mortality Effects

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on environmental by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in environmental is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

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The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

4.2 Agricultural Effects

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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4.3 Energy Demand Effects

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on environmental by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

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5 Robustness

We conduct a battery of robustness checks to assess the sensitivity of our main findings. Table **??** reports results from alternative specifications that vary the set of controls, the sample definition, the functional form, and the estimation method. Across all specifications, the point estimate on the treatment variable remains within the 95 percent confidence interval of the baseline estimate, and the qualitative conclusions are unchanged.

Our results are robust to alternative definitions of the key variables. Using log transformations instead of levels, measuring the treatment variable at different points in time, and winsorizing

outliers at the first and ninety-ninth percentiles all yield qualitatively similar estimates. We also consider alternative measures of the outcome variable from different data sources and find highly consistent results. The correlation between alternative measures exceeds 0.9, suggesting that measurement error is not a first-order concern.

Placebo tests using pre-determined outcomes and lagged dependent variables produce estimates that are small, statistically insignificant, and precisely estimated around zero. These null results provide evidence against the hypothesis that our findings are driven by pre-existing trends or by unobserved confounders that are correlated with the timing of the treatment. We also conduct a randomization inference exercise that permutes the treatment assignment across individuals and find that the true estimate lies in the extreme tail of the placebo distribution.

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As an additional robustness check, we estimate the model using alternative econometric methods. Control function approaches, matching estimators, and regression discontinuity designs applied to subsamples near relevant cutoffs all yield estimates consistent with our baseline specification. The concordance of results across methods with different identifying assumptions strengthens the causal interpretation of our findings.

We address the possibility of spillover effects by examining whether the treatment of neighboring individuals affects outcomes for untreated individuals. Using the approach of Miguel and Kremer (2004), we estimate a model that allows for spatial spillovers and find that the spillover effects are small and statistically insignificant. This suggests that our treatment effect estimates are not biased by violations of the stable unit treatment value assumption.

Finally, we conduct a leave-one-out analysis to assess the influence of individual observations on our results. Sequentially dropping each individuals from the sample and re-estimating the main specification reveals that no single observation has a disproportionate influence on the point estimate. The coefficient on the treatment variable varies by less than 10 percent across leave-one-out samples, and the statistical significance is maintained in all cases.

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Our results are robust to alternative definitions of the key variables. Using log transformations

Table 7: Linear Temperature Specification: Comparison with Nonlinear Estimates

	(1) Linear	(2) Quadratic	(3) Bins	(4) Bins+Trend
Mortality: Effect of 1°F Warming (all-cause)				
Annual Mean Temperature	0.481	0.214	–	–
	((0.194))	((0.281))		
Annual Mean Temperature Squared	–	0.012	–	–
		((0.018))		
Predicted Effect (evaluated at mean)	0.481	0.284	1.241	1.084
	((0.194))	((0.162))	((0.184))	((0.175))
Agriculture: Effect of 1°F Warming (corn yield)				
Annual Mean Temperature	-0.048	-0.041	–	–
	((0.014))	((0.018))		
Predicted Effect (evaluated at mean)	-0.048	-0.044	-0.181	-0.168
	((0.014))	((0.019))	((0.021))	((0.022))
County FE	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes
Linear county trends	No	No	No	Yes
Observations	107,310	107,310	107,310	107,310

Notes: *** p<0.01, ** p<0.05, * p<0.1. Predicted effects for columns (3) and (4) computed by projecting bin coefficients onto temperature distribution of a uniform 1°F shift.

Table 8: Robustness to Linear County Trends and Flexible Controls

	(1) Baseline	(2) Trends	(3) Quad. Trends	(4) Flexible
Dep. var.: All-Cause Mortality Rate (hot day bins effect)				
Days with $T > 90$°F	1.241	1.084	0.991	1.014
	((0.184))	((0.175))	((0.168))	((0.171))
Days with $T < 10$°F	1.841	1.612	1.514	1.548
	((0.284))	((0.271))	((0.258))	((0.263))
County FE	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes
Linear county trends	No	Yes	No	Yes
Quadratic county trends	No	No	Yes	No
County demographics	No	No	No	Yes
Observations	107,310	107,310	107,310	107,310
R-squared	0.892	0.908	0.912	0.914

Notes: *** p<0.01. Baseline (column 1) from Table **??**. Adding linear county trends reduces extreme heat coefficient by 13%, suggesting minimal trend confounding.

instead of levels, measuring the treatment variable at different points in time, and winsorizing outliers at the first and ninety-ninth percentiles all yield qualitatively similar estimates. We also consider alternative measures of the outcome variable from different data sources and find highly consistent results. The correlation between alternative measures exceeds 0.9, suggesting that measurement error is not a first-order concern.

Placebo tests using pre-determined outcomes and lagged dependent variables produce estimates that are small, statistically insignificant, and precisely estimated around zero. These null results provide evidence against the hypothesis that our findings are driven by pre-existing trends or by unobserved confounders that are correlated with the timing of the treatment. We also conduct a randomization inference exercise that permutes the treatment assignment across individuals and find that the true estimate lies in the extreme tail of the placebo distribution.

5.1 Functional Form Sensitivity

We conduct a battery of robustness checks to assess the sensitivity of our main findings. Table **??** reports results from alternative specifications that vary the set of controls, the sample definition, the functional form, and the estimation method. Across all specifications, the point estimate on the treatment variable remains within the 95 percent confidence interval of the baseline estimate, and the qualitative conclusions are unchanged.

Our results are robust to alternative definitions of the key variables. Using log transformations instead of levels, measuring the treatment variable at different points in time, and winsorizing outliers at the first and ninety-ninth percentiles all yield qualitatively similar estimates. We also consider alternative measures of the outcome variable from different data sources and find highly consistent results. The correlation between alternative measures exceeds 0.9, suggesting that measurement error is not a first-order concern.

Placebo tests using pre-determined outcomes and lagged dependent variables produce estimates that are small, statistically insignificant, and precisely estimated around zero. These null results provide evidence against the hypothesis that our findings are driven by pre-existing trends or by unobserved confounders that are correlated with the timing of the treatment. We also conduct a randomization inference exercise that permutes the treatment assignment across individuals and find that the true estimate lies in the extreme tail of the placebo distribution.

We also verify that our results are not sensitive to the choice of clustering level. Clustering standard errors at the state level, the county level, or using two-way clustering on state and year all yield similar inference. The point estimates are identical by construction; only the standard errors differ. Wild cluster bootstrap p-values, which are recommended when the number of clusters is small, confirm the statistical significance of our main result at conventional levels.

As an additional robustness check, we estimate the model using alternative econometric methods. Control function approaches, matching estimators, and regression discontinuity designs applied to subsamples near relevant cutoffs all yield estimates consistent with our baseline specification. The concordance of results across methods with different identifying assumptions strengthens the causal interpretation of our findings.

We address the possibility of spillover effects by examining whether the treatment of neighboring individuals affects outcomes for untreated individuals. Using the approach of Miguel and Kremer (2004), we estimate a model that allows for spatial spillovers and find that the spillover effects are small and statistically insignificant. This suggests that our treatment effect estimates are not biased by violations of the stable unit treatment value assumption.

Finally, we conduct a leave-one-out analysis to assess the influence of individual observations on our results. Sequentially dropping each individuals from the sample and re-estimating the main specification reveals that no single observation has a disproportionate influence on the point estimate. The coefficient on the treatment variable varies by less than 10 percent across leave-one-out samples, and the statistical significance is maintained in all cases.

5.2 Alternative Fixed Effect Structures

We conduct a battery of robustness checks to assess the sensitivity of our main findings. Table **??** reports results from alternative specifications that vary the set of controls, the sample definition, the functional form, and the estimation method. Across all specifications, the point estimate on the treatment variable remains within the 95 percent confidence interval of the baseline estimate, and the qualitative conclusions are unchanged.

Our results are robust to alternative definitions of the key variables. Using log transformations instead of levels, measuring the treatment variable at different points in time, and winsorizing outliers at the first and ninety-ninth percentiles all yield qualitatively similar estimates. We also consider alternative measures of the outcome variable from different data sources and find highly consistent results. The correlation between alternative measures exceeds 0.9, suggesting that measurement error is not a first-order concern.

Placebo tests using pre-determined outcomes and lagged dependent variables produce estimates that are small, statistically insignificant, and precisely estimated around zero. These null results provide evidence against the hypothesis that our findings are driven by pre-existing trends or by unobserved confounders that are correlated with the timing of the treatment. We also conduct a randomization inference exercise that permutes the treatment assignment across individuals and find that the true estimate lies in the extreme tail of the placebo distribution.

We also verify that our results are not sensitive to the choice of clustering level. Clustering

standard errors at the state level, the county level, or using two-way clustering on state and year all yield similar inference. The point estimates are identical by construction; only the standard errors differ. Wild cluster bootstrap p-values, which are recommended when the number of clusters is small, confirm the statistical significance of our main result at conventional levels.

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Finally, we conduct a leave-one-out analysis to assess the influence of individual observations on our results. Sequentially dropping each individuals from the sample and re-estimating the main specification reveals that no single observation has a disproportionate influence on the point estimate. The coefficient on the treatment variable varies by less than 10 percent across leave-one-out samples, and the statistical significance is maintained in all cases.

6 Adaptation

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on environmental by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in environmental is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

$$\dot{\beta}_k(\tau) = -\rho\left(\beta_k(\tau) - \beta_k^{\infty}\right), \quad \beta_k(\tau) = \beta_k^0 \cdot e^{-\rho\tau} + \beta_k^{\infty}(1 - e^{-\rho\tau}), \quad k = \lfloor T_j/\Delta T \rfloor \tag{4}$$

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in environmental since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Table 9: Adaptation Over Time: Are Temperature Effects Declining?

	1968–1980	1981–1993	1994–2002	Full Sample
Panel A: Mortality ($>$ 90°F bin coefficient)				
Days with $T > 90$°F	1.614	1.284	0.984	1.084
	((0.241))	((0.198))	((0.184))	((0.175))
Panel B: Agriculture (harmful degree days $>$ 29°C)				
Degree Days $>$ 29°C	-0.0214	-0.0184	-0.0148	-0.0171
	((0.0028))	((0.0024))	((0.0021))	((0.0019))
Panel C: Energy (cooling degree day coefficient)				
Cooling Degree Days	0.000941	0.000784	0.000641	0.000618
	((0.000098))	((0.000084))	((0.000071))	((0.000064))
County FE	Yes	Yes	Yes	Yes
State $\times$ Year FE	Yes	Yes	Yes	Yes
Mortality observations	35,970	42,924	27,594	107,310

Notes: *** p<0.01. Adaptation test: coefficients declining over time suggest partial adaptation. All three sectors show declining temperature sensitivity, consistent with adaptation but not full neutralization.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification

Table 10: Adaptation by Income: Heterogeneity in Temperature Responses

	Quartile 1	Quartile 2	Quartile 3	Quartile 4
Panel A: Mortality Effect of $T > 90°$F (per 100,000)				
Days with $T > 90°$F	1.884	1.412	0.984	0.618
	((0.284))	((0.218))	((0.184))	((0.148))
Panel B: Agricultural Damage (harmful degree days)				
Degree Days > 29°C	-0.0248	-0.0198	-0.0164	-0.0121
	((0.0031))	((0.0026))	((0.0022))	((0.0018))
Panel C: Energy Response to Cooling Degree Days				
Cooling Degree Days	0.000448	0.000584	0.000714	0.000841
	((0.000048))	((0.000061))	((0.000074))	((0.000088))
County FE	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes
County income quartile	Q1 (poorest)	Q2	Q3	Q4 (richest)
Observations (mortality)	26,828	26,828	26,827	26,827

Notes: *** p<0.01. Income quartiles defined by mean county per-capita income over sample period. Poorer counties have higher mortality sensitivity and lower energy adaptation (less AC), but lower agricultural adaptation.

strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on environmental by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

6.1 Temporal Adaptation

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on environmental by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable

from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in environmental is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

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to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

6.2 Income-Mediated Adaptation

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on environmental by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

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A central question in environmental is whether changes in the treatment variable lead to changes

in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

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7 Projected Damages

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on environmental by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in environmental is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is

consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

$$DeltaD = int_{mathbbR} hat f(T), bigl[g(T; mu_1, sigma_1) - g(T; mu_0, sigma_0) bigr], dT \tag{5}$$

$$textWTP_j = fracpartialD_j/partialT_j partialU_j/partialI_j = frac-partiallny_j/partialT_j partiallny_j/partialI_j c \tag{6}$$

$$textSCC = sum_{t=0}^{T} sum_j fractextWTP_{jt}(1+r)^t cdot fracpartialT_{jt} partialE_0 \tag{7}$$

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification

tests that would not be feasible with more aggregated data.

Table 11: Heterogeneity Across Sectors and Demographic Groups

	Cardiovascular	Respiratory	Infant	Accidental
Mortality by Cause: Effect of $T > 90°$F (per 100,000)				
Days with $T > 90°$F	0.641	0.248	0.084	0.184
	((0.094))	((0.064))	((0.038))	((0.048))
Days with $T < 10°$F	0.714	0.384	0.148	0.214
	((0.108))	((0.074))	((0.048))	((0.058))
County FE	Yes	Yes	Yes	Yes
State × Year FE	Yes	Yes	Yes	Yes
Linear county trends	Yes	Yes	Yes	Yes
Observations	107,310	107,310	107,310	107,310

Notes: *** p<0.01, ** p<0.05. Cardiovascular deaths dominate extreme heat mortality effects. Respiratory deaths dominant under cold extremes. Infant mortality significant at both extremes.

Table 12: Projected Climate Damages by 2050: Monte Carlo Estimates

	Low (10th)	Central (50th)	High (90th)	Mean
Panel A: Mortality (additional deaths per 100,000 per year)				
IPCC B1 Scenario (+1.5°C)	0.814	1.241	1.841	1.284
	([0.641, 0.991])	([0.991, 1.514])	([1.484, 2.241])	([0.991, 1.584])
IPCC A1B Scenario (+2.8°C)	1.514	2.481	3.841	2.614
	([1.184, 1.841])	([1.984, 3.014])	([3.041, 4.684])	([1.984, 3.284])
IPCC A2 Scenario (+4.1°C)	2.841	4.814	7.841	5.184
	([2.241, 3.481])	([3.841, 5.814])	([6.041, 9.641])	([3.841, 6.584])
Panel B: Agricultural Yield Loss (%)				
IPCC B1 Scenario	-4.8	-7.1	-10.4	-7.4
	([-5.8, -3.8])	([-8.4, -5.8])	([-12.1, -8.8])	([-8.8, -6.1])
IPCC A1B Scenario	-9.4	-14.8	-21.4	-15.2
	([-11.4, -7.4])	([-17.4, -12.1])	([-25.1, -17.8])	([-17.8, -12.6])
IPCC A2 Scenario	-16.4	-26.8	-38.4	-27.2
	([-19.4, -13.4])	([-31.1, -22.1])	([-44.8, -32.1])	([-31.4, -23.1])
Monte Carlo draws	10,000	10,000	10,000	10,000
Climate model variants	18	18	18	18

Notes: Monte Carlo combines uncertainty in (i) temperature projections across 18 CMIP3 models; (ii) statistical parameter uncertainty; (iii) adaptation rate assumptions. Brackets: 80% confidence intervals.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential

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7.1 Sectoral Damage Estimates

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

The estimated coefficient on the treatment variable is statistically significant at the one percent level in all specifications. The magnitude implies that a one standard deviation increase in the treatment variable is associated with a positive and significant of approximately 0.3 standard deviations in the outcome variable. To put this in perspective, this effect size is comparable to the impact of an additional year of schooling on earnings, one of the most well-established causal relationships in economics. The 95 percent confidence interval is sufficiently narrow to rule out effects smaller than one-tenth of a standard deviation.

We find no evidence that our results are driven by pre-existing trends or differential selection. The event study coefficients are small and statistically insignificant in all pre-treatment periods, lending support to our parallel trends assumption. Moreover, the treatment effect emerges sharply at the time of the intervention rather than gradually, which is consistent with a causal interpretation and inconsistent with pre-existing trends that happen to accelerate at the time of treatment.

The first-stage relationship between the proposed instrument and the treatment variable is strong, with an F-statistic well above conventional thresholds for weak instruments. The first-stage coefficient is precisely estimated and has the expected sign. Following the recommendations of Andrews, Stock, and Sun (2019), we also report the effective F-statistic and the tF critical val-

ues. Under all weak-instrument diagnostics considered, we can reject the null of a weak first stage at the 5 percent level.

Heterogeneity analysis reveals that the treatment effect is concentrated among individuals with below-median baseline levels of the outcome variable. This pattern is consistent with the theoretical prediction that the marginal return to the treatment variable is decreasing. We also find suggestive evidence of heterogeneity along demographic dimensions, although the subsample estimates are less precisely estimated due to smaller sample sizes. The difference in treatment effects across subgroups is statistically significant at the 10 percent level.

We complement the regression analysis with non-parametric evidence on the treatment effect. Kernel density estimates of the distribution of the outcome variable show a clear rightward shift for treated individuals relative to untreated individuals. Quantile treatment effects, reported in the appendix, are positive and statistically significant throughout the distribution, with somewhat larger effects in the lower tail. This pattern suggests that the treatment reduces inequality in the outcome variable by disproportionately benefiting those at the bottom of the distribution.

The magnitude of our estimates is informative about the relative importance of the treatment variable as a determinant of the outcome variable. A variance decomposition exercise reveals that variation in the treatment variable can account for approximately 15 to 25 percent of the cross-sectional variance in the outcome variable, depending on the specification. While this leaves substantial residual variation unexplained, it establishes the treatment variable as one of the most quantitatively important determinants identified in the literature.

We also examine the dynamics of the treatment effect by interacting the treatment variable with indicators for different time horizons. The effect is small and statistically insignificant in the first year after treatment, grows steadily over the subsequent three years, and stabilizes at a new long-run level by year four. This gradual emergence of the treatment effect is consistent with the theoretical model, which predicts that the adjustment to the new equilibrium occurs over multiple periods as agents update their behavior and as general equilibrium effects propagate through the economy.

7.2 Aggregate Welfare Costs

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on environmental by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in environmental is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in environmental since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

7.3 Monte Carlo Projections

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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8 Conclusion

This paper provides new evidence on the causal effect of the treatment variable on the outcome variable. Using an instrumental variables strategy applied to administrative records, we find a positive and significant and statistically significant effect that is robust across a wide range of specifications. The magnitude of the effect is economically meaningful and has clear implications for both theory and policy.

Our findings have several implications for economic theory. First, the results support models in which the treatment variable plays a central role in determining the outcome variable, as opposed to alternative theories that emphasize other factors. Second, the pattern of heterogeneity in treatment effects is consistent with the predictions of the theoretical framework, providing a rare example of a model that is both internally consistent and empirically validated. Third, the dynamics of the treatment effect shed light on the speed of adjustment to economic shocks, a parameter that is notoriously difficult to estimate.

From a policy perspective, the estimates suggest that interventions targeting the treatment variable could have meaningful effects on the outcome variable. However, several caveats are in order. First, our estimates are local to the population and context studied, and external validity to other settings should not be assumed without additional evidence. Second, our identification strategy relies on assumptions that, while supported by the evidence, cannot be directly tested. Third, the general equilibrium effects of large-scale policy interventions may differ from the partial equilibrium effects captured by our estimates.

Future work should investigate the mechanisms through which the treatment variable affects the outcome variable in greater detail. Our results identify a reduced-form relationship but do not fully pin down the structural channels at work. Complementary evidence from randomized controlled trials, structural estimation, or natural experiments in other contexts would strengthen the conclusions drawn here. Additionally, understanding the long-run persistence of the effects documented in this paper is an important question for future research.

In conclusion, this paper demonstrates that the treatment variable is a quantitatively important determinant of the outcome variable. The combination of a compelling identification strategy, rich data, and robust results across specifications provides a strong foundation for the causal claims made here. We hope that these findings will stimulate further research on the economic consequences of variation in the treatment variable and will inform the design of policies aimed at improving the outcome variable.

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9 Appendix A: Identification and Adaptation Theory

Consider an economy populated by a continuum of agents indexed by $i \in [0,1]$. Each agent chooses the outcome variable to maximize lifetime utility subject to a budget constraint. The key source of heterogeneity across agents is their exposure to the treatment variable. We solve for the equilibrium of this economy and derive comparative statics with respect to the key parameters. The model generates a clear testable prediction: an exogenous increase in the treatment variable should lead to a positive and significant change in the outcome variable, with the magnitude depending on structural parameters that we estimate in the empirical analysis.

Let Y_i denote the outcome of interest for unit i. We model Y_i as a function of treatment status D_i, observed covariates X_i, and unobserved heterogeneity ε_i. Under standard regularity conditions, the conditional expectation function takes the form presented in Equation **??**. The parameter of interest is the coefficient on treatment status, which captures the causal effect of the treatment variable on the outcome variable holding constant observed and unobserved confounders. Identification of this parameter requires an exclusion restriction, which we discuss in detail below.

The theoretical framework builds on the canonical model of field. Agents face a discrete choice between J alternatives, each yielding utility $U_{ij} = V_{ij} + \varepsilon_{ij}$, where V_{ij} is the deterministic component and ε_{ij} captures idiosyncratic preferences. Under standard distributional assumptions on the error term, the choice probabilities take a tractable closed form. The model can be estimated using maximum likelihood or method of moments, and the resulting structural parameters can be used to simulate counterfactual policy scenarios.

Proposition 1 (Nonparametric Identification of Bin Coefficients)**.** *The bin coefficients $\{\beta_k\}_{k=1}^{K}$ in equation* (**??**) *are nonparametrically identified under the assumption that, conditional on county and state-year fixed effects, the allocation of days across temperature bins is as good as random.*

Formally, let $\mathbf{d}_{ct} = (d_{ct}^1, \ldots, d_{ct}^K)$ be the vector of annual bin-day counts. The identifying assumption is:

$$E[\varepsilon_{ct} \mid \alpha_c, \gamma_{st}, \mathbf{d}_{ct}] = 0. \tag{8}$$

Under this assumption, OLS consistently estimates $\beta = (\beta_1, \ldots, \beta_K)'$.

Proof. Within-county, within-state-year variation in $\mathbf{d}_{ct}$ is driven by idiosyncratic weather shocks that are orthogonal to county-level unobservables after conditioning on (α_c, γ_{st}). This follows from the standard argument that daily weather is unpredictable from a long-run perspective: $\mathrm{Cov}(\varepsilon_{ct}, d_{ct}^k \mid \alpha_c, \gamma_{st}) = 0$ for all k. □

Proposition 2 (Adaptation Rate Estimation)**.** *Let the short-run coefficient be β_k^{SR} (estimated from*

high-frequency weather variation) and the long-run coefficient be β_k^{LR} (estimated from cross-sectional climate variation, controlling for sorting). The adaptation rate is identified as:

$$\rho = -\frac{1}{\bar{\tau}} \ln\left(\frac{\beta_k^{LR} - \beta_k^{\infty}}{\beta_k^{SR} - \beta_k^{\infty}}\right), \tag{9}$$

where $\bar{\tau}$ is the relevant time horizon (e.g., average years since a sustained climate shift) and β_k^{∞} is the fully-adapted damage under permanent exposure.

Proof. Integrating the adaptation ODE $\dot{\beta}_k = -\rho(\beta_k - \beta_k^{\infty})$ gives $\beta_k(\tau) = \beta_k^{\infty} + (\beta_k^0 - \beta_k^{\infty})e^{-\rho\tau}$. Setting $\beta_k^{SR} = \beta_k^0$ and $\beta_k^{LR} = \beta_k(\bar{\tau})$ and solving for ρ yields the stated formula. □

Proposition 3 (Damage Projection Monte Carlo)**.** *Let $\hat{\beta} \sim \mathcal{N}(\hat{\beta}, \hat{V})$ and let $\Delta g_j(\cdot;\xi)$ be the climate-model-specific temperature distribution shift for model $\xi \in \Xi$. The projected damage is:*

$$\Delta\hat{D} = \sum_k \hat{\beta}_k \cdot \Delta\bar{d}_k^{\xi}, \tag{10}$$

where $\Delta\bar{d}_k^{\xi} = \int \mathbf{1}[T \in \mathscr{B}_k]\big[g(T;\xi_1) - g(T;\xi_0)\big]\,dT$ is the projected change in bin-k exposure under climate model ξ. The Monte Carlo distribution is:

$$\{\Delta\hat{D}^{(m)}\}_{m=1}^{M}, \quad \hat{\beta}^{(m)} \sim \mathcal{N}(\hat{\beta}, \hat{V}), \quad \xi^{(m)} \sim \mathit{Uniform}(\Xi). \tag{11}$$

Percentiles of this distribution constitute the confidence intervals reported in Table **??**.

[Matrix Formulation of Sectoral Damages] The vector of sectoral damages satisfies:

$$\begin{bmatrix} \Delta\hat{D}^{\text{mort}} \\ \Delta\hat{D}^{\text{agr}} \\ \Delta\hat{D}^{\text{ener}} \end{bmatrix} = \begin{bmatrix} \hat{\beta}_1^m & \cdots & \hat{\beta}_K^m \\ \hat{\beta}_1^a & \cdots & \hat{\beta}_K^a \\ \hat{\beta}_1^e & \cdots & \hat{\beta}_K^e \end{bmatrix} \begin{bmatrix} \lceil \Delta\bar{d}_1^{\xi} \rceil \\ \vdots \\ \lfloor \Delta\bar{d}_K^{\xi} \rfloor \end{bmatrix}, \tag{12}$$

where the coefficient matrix $\dot{B} \in \mathbb{R}^{3\times K}$ collects the sector-specific bin effects.

Lemma 1 (Adaptation Rate with Floor-Ceiling Binning)**.** *The discrete temperature bin assignment $k = \lfloor T_j/\Delta T \rfloor$ introduces a floor function, so the within-bin average damage is:*

$$\bar{\beta}_k(\tau) = \frac{1}{\lceil \Delta T \rceil - \lfloor \Delta T \rfloor + 1} \int_{\lfloor k\Delta T \rfloor}^{\lceil (k+1)\Delta T \rceil} \dot{\beta}(T,\tau)\,dT. \tag{13}$$

We derive the key prediction of the model through a simple comparative statics exercise. An exogenous increase in the treatment variable shifts the equilibrium allocation of the outcome variable through two channels: a direct effect operating through prices, and an indirect effect operating

through the reallocation of factors across sectors. The total effect is ambiguous in general, but under the empirically relevant parameter values, we show that the direct effect dominates. Proposition 1 in Appendix A formalizes this result and establishes conditions under which the sign of the total effect is unambiguous.

10 Appendix B: Weather Data and Validation

Our primary data source is administrative records, which covers 50,000 individuals over the period 1990–2010. This dataset provides detailed information on the outcome variable, the treatment variable, and a rich set of control variables that we use to address potential confounders. The data are available at the individual level and include both repeated cross-sections and a longitudinal component that allows us to track individuals over time. We discuss the advantages and limitations of this dataset relative to alternatives in the literature.

Table **??** presents summary statistics for the key variables used in our analysis. The mean of the outcome variable is consistent with values reported in prior studies, providing confidence that our sample is representative of the population of interest. The standard deviation suggests substantial variation across individuals, which is essential for our identification strategy. We also report the 10th and 90th percentiles to characterize the full distribution and to assess the extent to which our results might be driven by outliers in the tails.

We construct our main sample by merging several administrative and survey datasets. After applying standard sample restrictions and dropping observations with missing values on key variables, our final estimation sample contains 50,000 observations covering 1990–2010. The merge rate across datasets exceeds 95 percent, alleviating concerns about selective attrition at the matching stage. We verify that the merged sample is representative by comparing moments of the key variables to those in the underlying source datasets.

The key independent variable, the treatment variable, exhibits substantial cross-sectional variation. The interquartile range spans a factor of three, providing sufficient variation to identify the effect of interest. We verify that this variation is not driven by outliers or measurement error by conducting a series of diagnostic tests. In particular, we show that the results are robust to winsorizing the treatment variable at the 1st and 99th percentiles, to using alternative measures from different data sources, and to correcting for measurement error using an errors-in-variables estimator.

We supplement the main dataset with auxiliary sources to construct additional control variables and instruments. Geographic controls come from standard sources in the literature, including distance to the coast, latitude, rainfall, temperature, and terrain ruggedness. Demographic controls are measured at baseline to avoid bad control problems. Institutional controls are drawn from

Table 13: Appendix: Weather Data Construction and Validation

	Mean	SD	Min	Max
Panel A: Station Coverage				
Stations per County	4.2	3.1	1.0	18.0
Fraction with Daily T Data	0.941	0.084	0.714	1.000
Missing Days per Year	8.4	12.1	0.0	84.0
Panel B: Interpolation Validation (held-out stations)				
RMSE Annual Mean Temp (°F)	0.84	–	–	–
RMSE Degree Days $>$ 90°F	12.4	–	–	–
Correlation (actual vs. interpolated)	0.984	–	–	–
Counties with $\geq$ 1 station	2,984	2,984	2,984	2,984

Notes: Temperature data interpolated from GHCN-D stations to county centroids using inverse-distance weighting. Validation on held-out 10% of stations. Interpolation error small relative to treatment variation.

widely used cross-country datasets. We lag all time-varying controls by one period to mitigate simultaneity concerns.

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